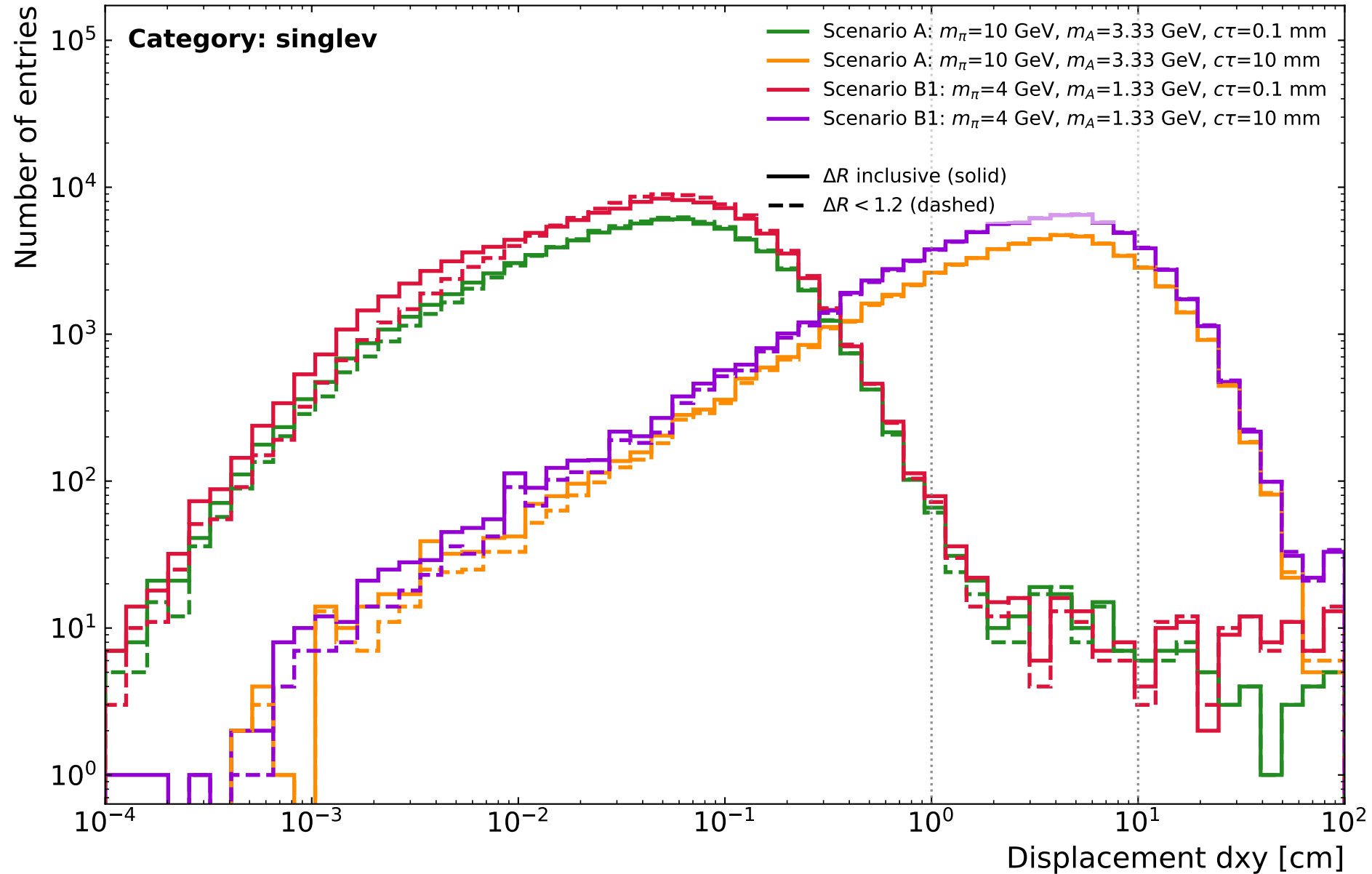


singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary

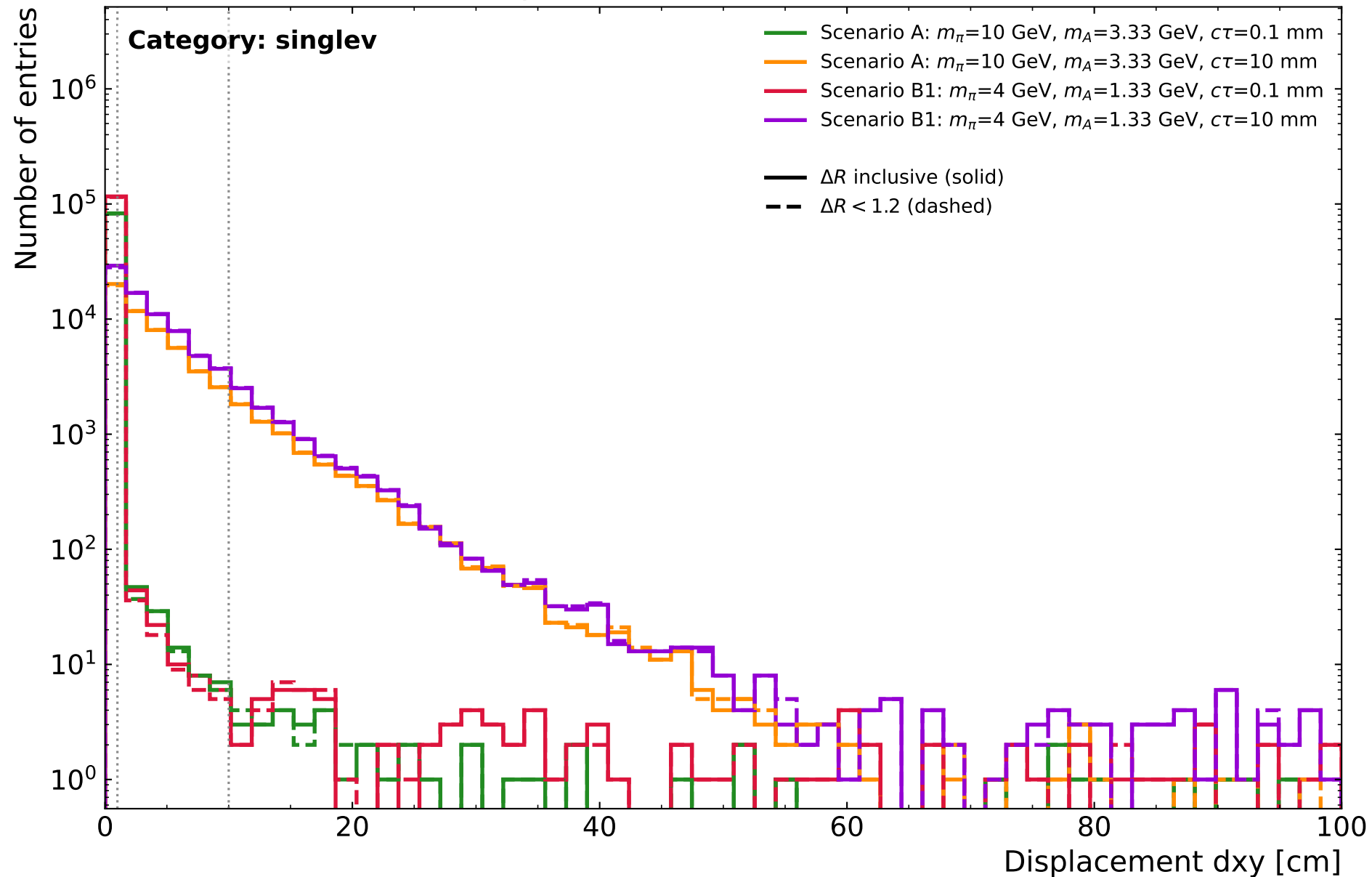
(13.6 TeV, 2024)



singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

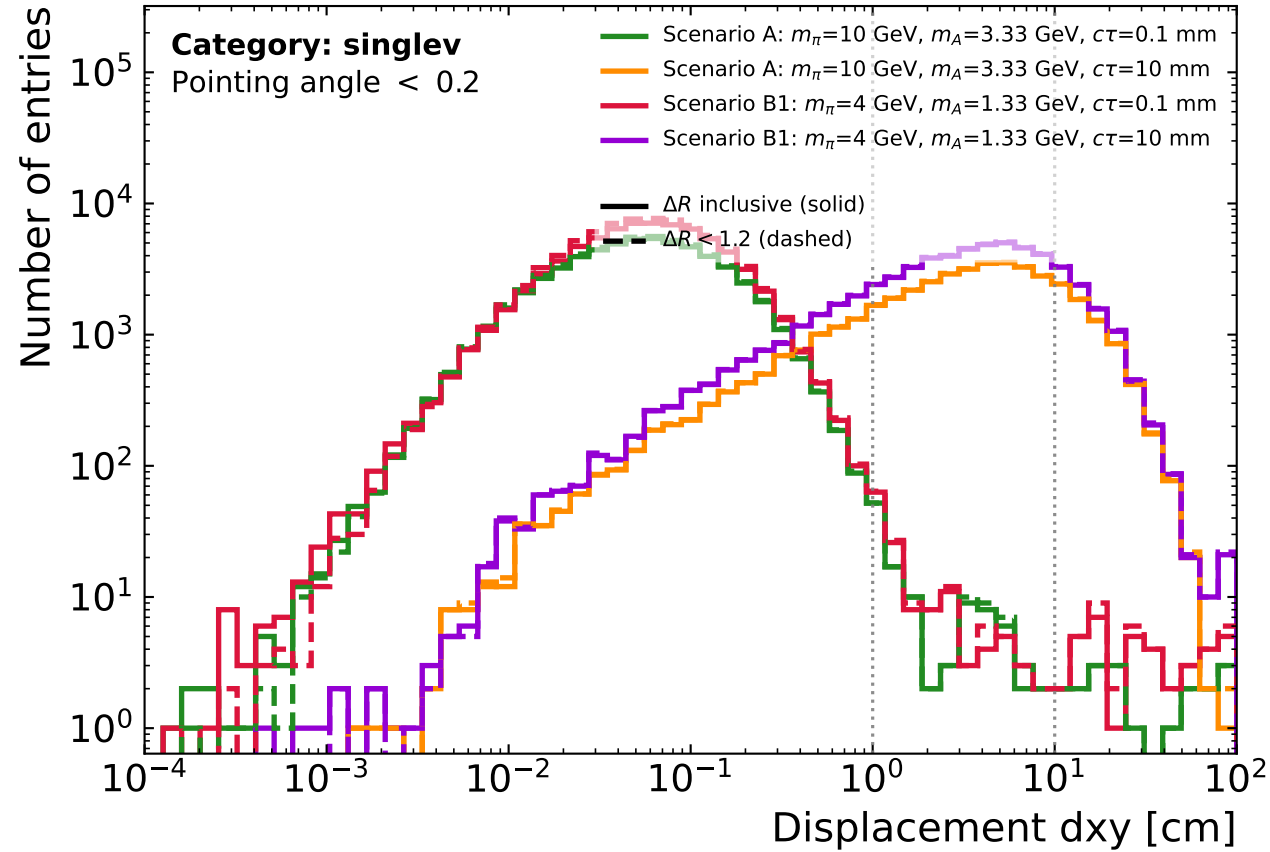
CMS Simulation Preliminary

(13.6 TeV, 2024)



singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

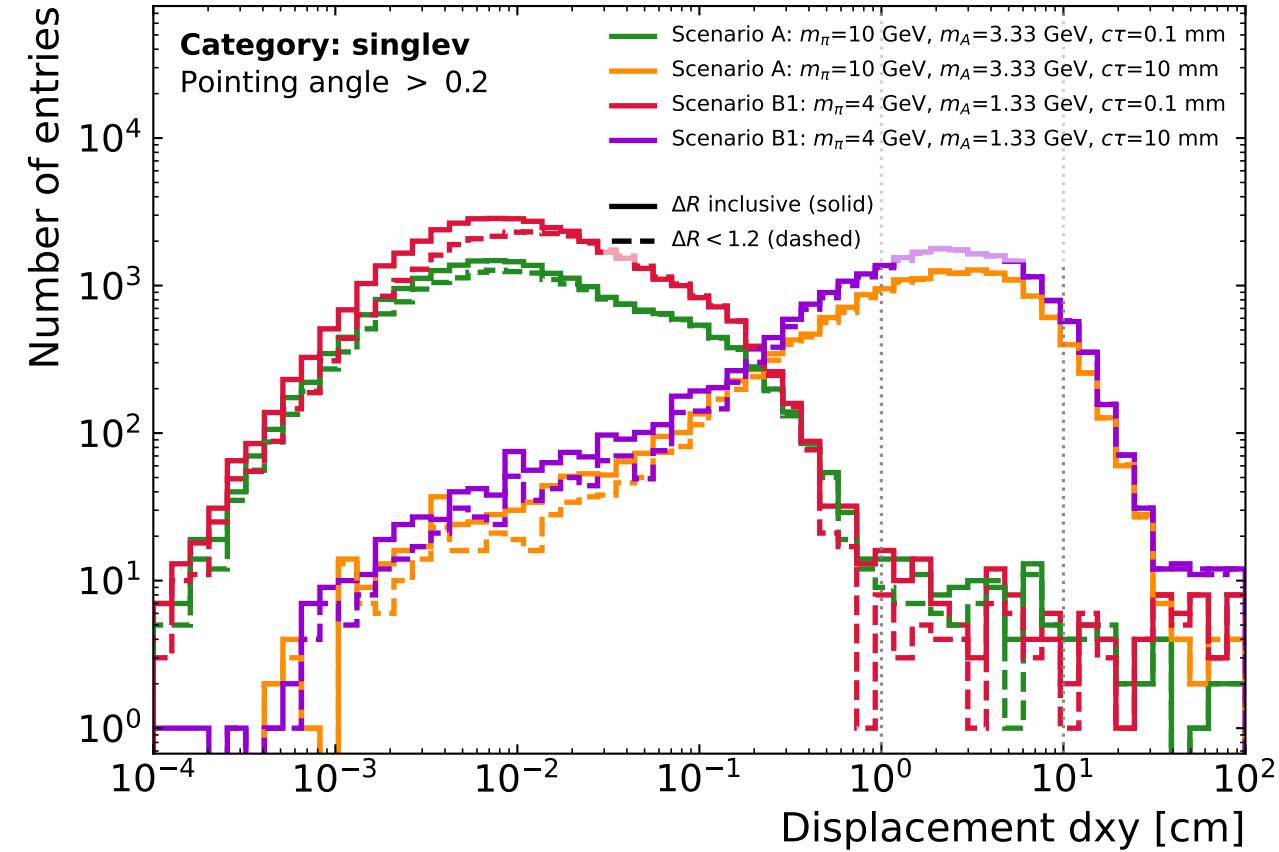
CMS Simulation Preliminary (13.6 TeV, 2024)



cell: $N(\Delta R \text{ inclusive})$ | $N(\Delta R < 1.2)$

	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	60,450 62,621	85 89	18 18	60,553 62,728
scA $c\tau=10$	8,169 8,231	27,401 27,603	6,641 6,701	42,211 42,535
scB1 $c\tau=0.1$	75,257 81,039	103 109	41 47	75,401 81,195
scB1 $c\tau=10$	11,847 11,943	39,135 39,543	8,457 8,557	59,439 60,043

CMS Simulation Preliminary (13.6 TeV, 2024)



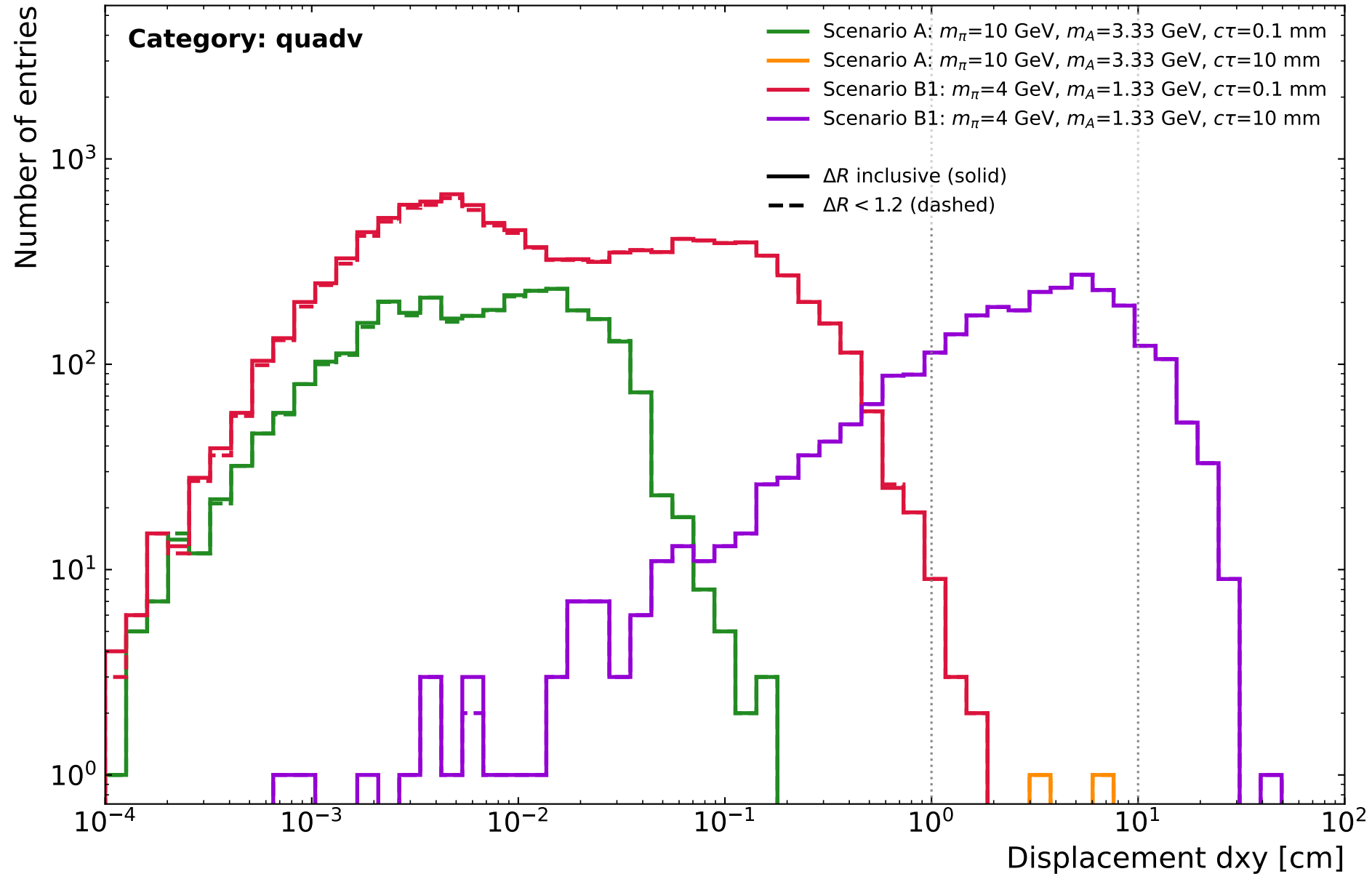
cell: $N(\Delta R \text{ inclusive})$ | $N(\Delta R < 1.2)$

	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	22,371 19,811	92 65	27 27	22,490 19,903
scA $c\tau=10$	5,347 4,774	10,450 10,367	812 812	16,609 15,953
scB1 $c\tau=0.1$	41,004 33,815	80 42	50 48	41,134 33,905
scB1 $c\tau=10$	7,600 6,601	14,441 14,313	1,133 1,138	23,174 22,052

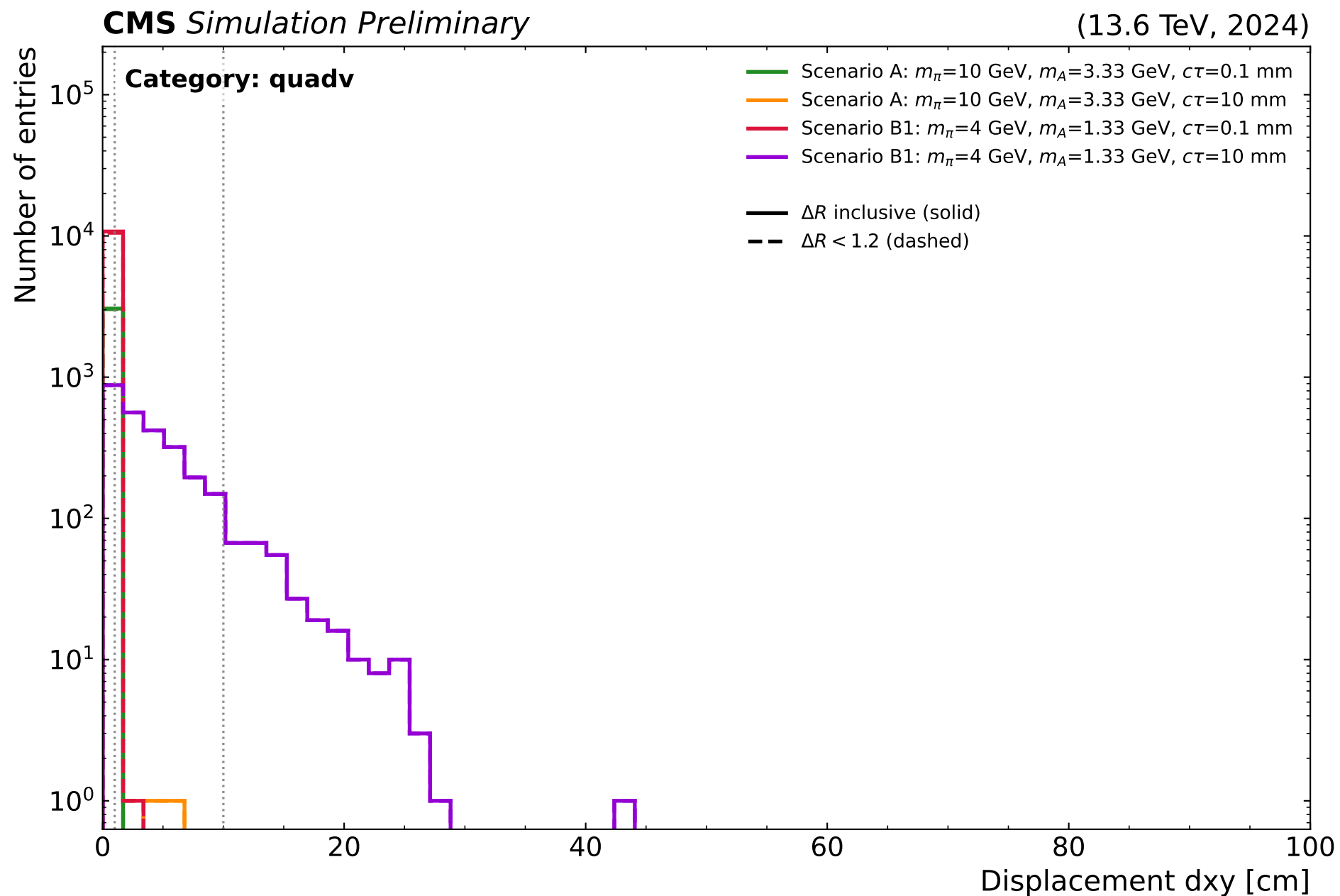
quadv: displacement of the binning vertex (fourmuonSV_dxy)
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary

(13.6 TeV, 2024)

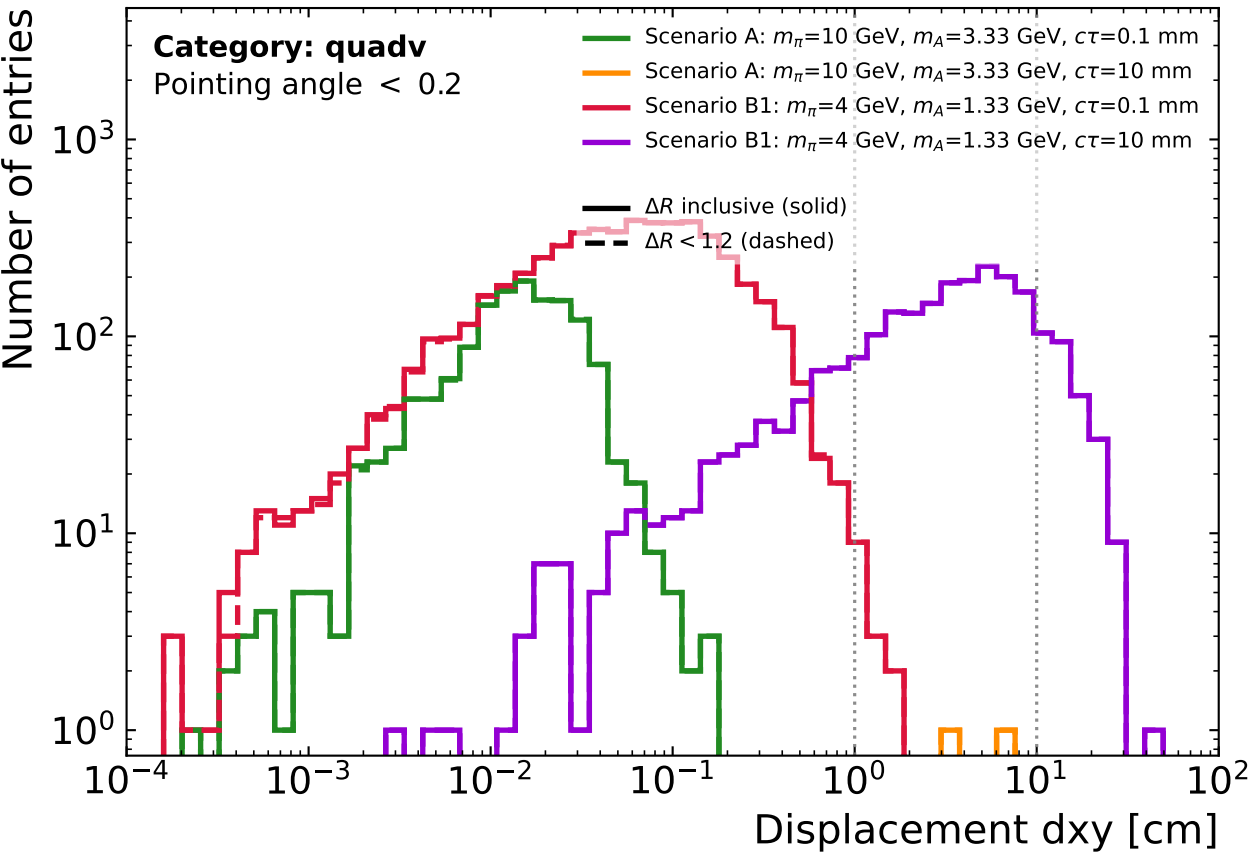


quadv: displacement of the binning vertex (fourmuonSV_dxy)
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)



quadv: displacement of the binning vertex (fourmuonSV_dxy), split by pointing angle
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

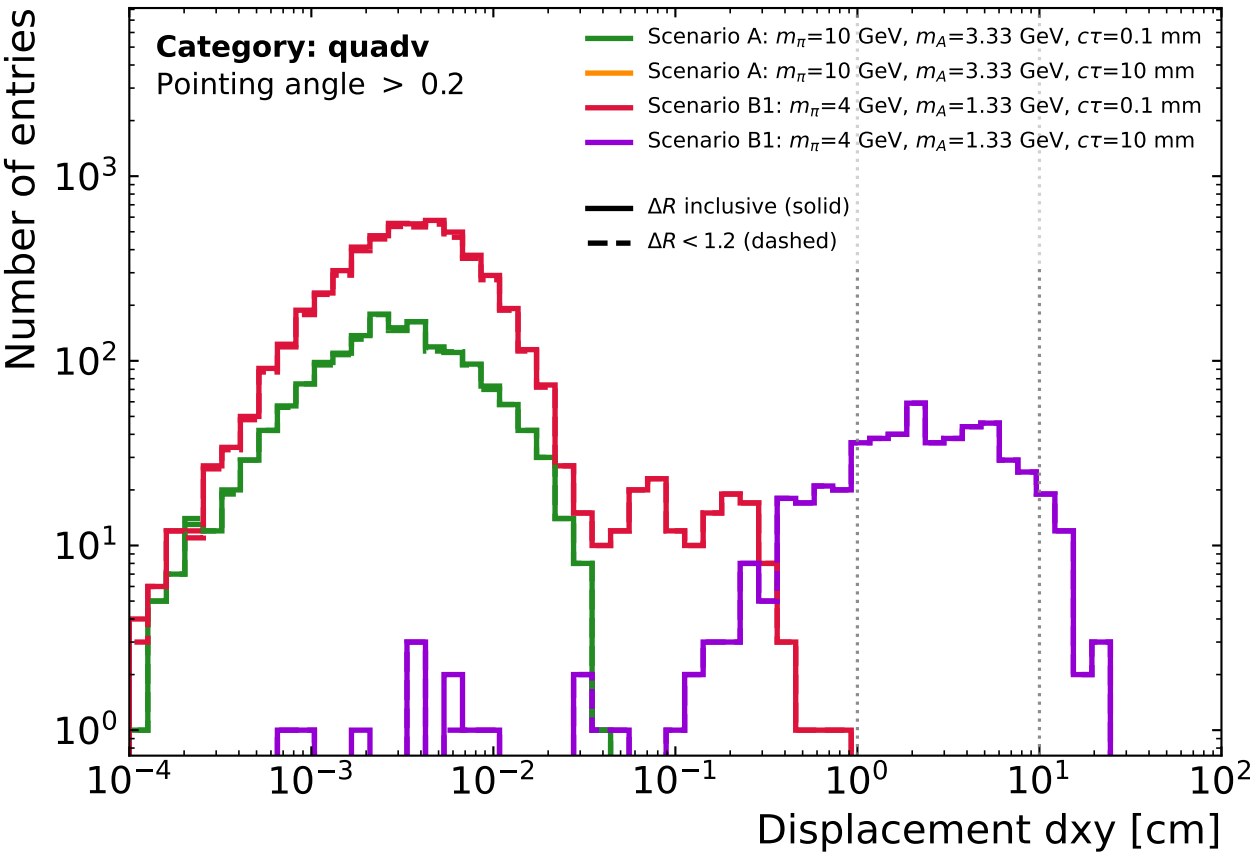
CMS Simulation Preliminary (13.6 TeV, 2024)



cell: $N(\Delta R \text{ inclusive}) \mid N(\Delta R < 1.2)$

	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	1,403 1,402	0 0	0 0	1,403 1,402
scA $c\tau=10$	0 0	2 2	0 0	2 2
scB1 $c\tau=0.1$	5,338 5,332	11 11	0 0	5,349 5,343
scB1 $c\tau=10$	449 449	1,553 1,553	267 267	2,269 2,269

CMS Simulation Preliminary (13.6 TeV, 2024)



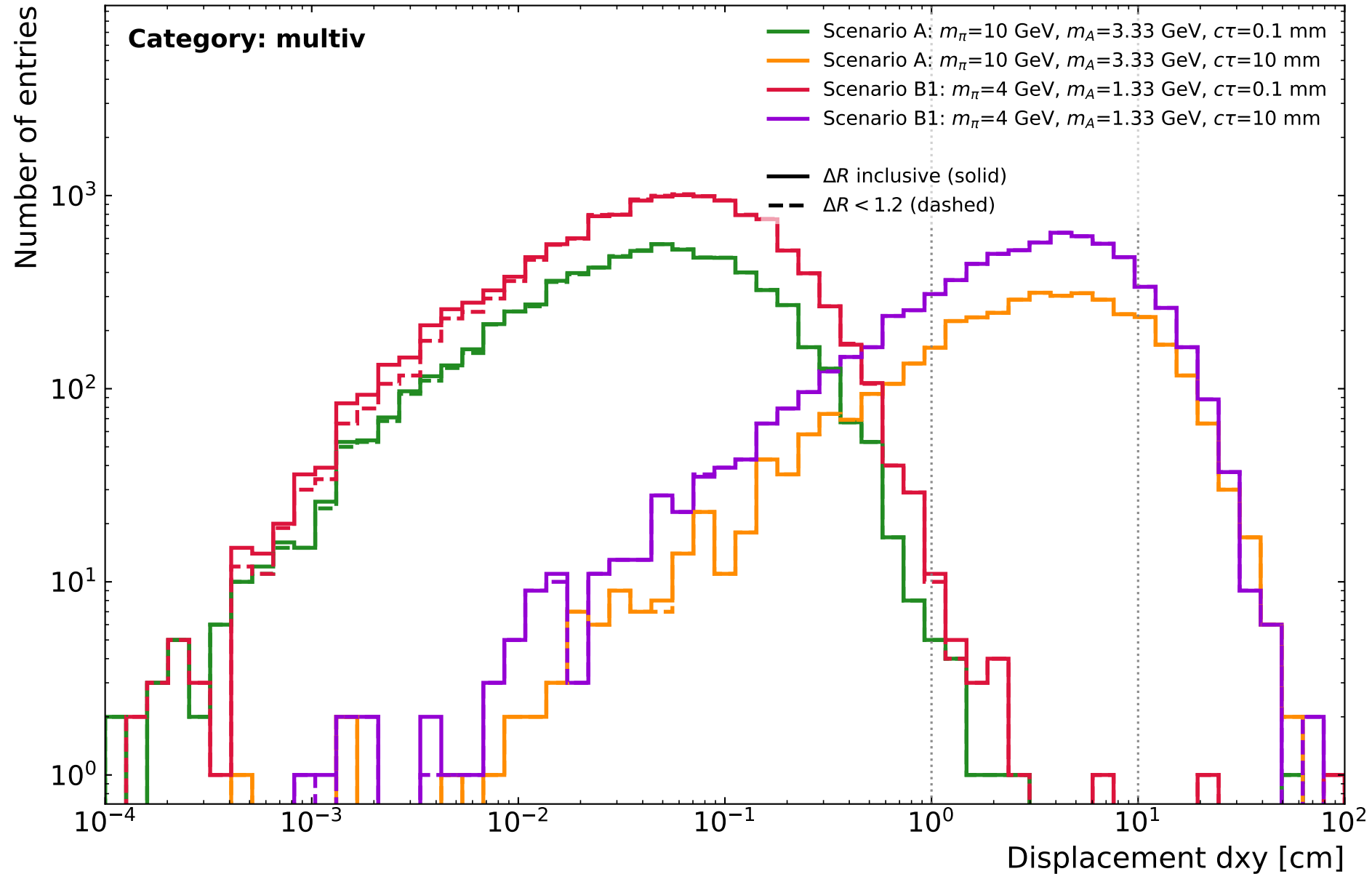
cell: $N(\Delta R \text{ inclusive}) \mid N(\Delta R < 1.2)$

	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	1,656 1,629	0 0	0 0	1,656 1,629
scA $c\tau=10$	0 0	0 0	0 0	0 0
scB1 $c\tau=0.1$	5,402 5,184	0 0	0 0	5,402 5,184
scB1 $c\tau=10$	123 121	386 386	30 30	539 537

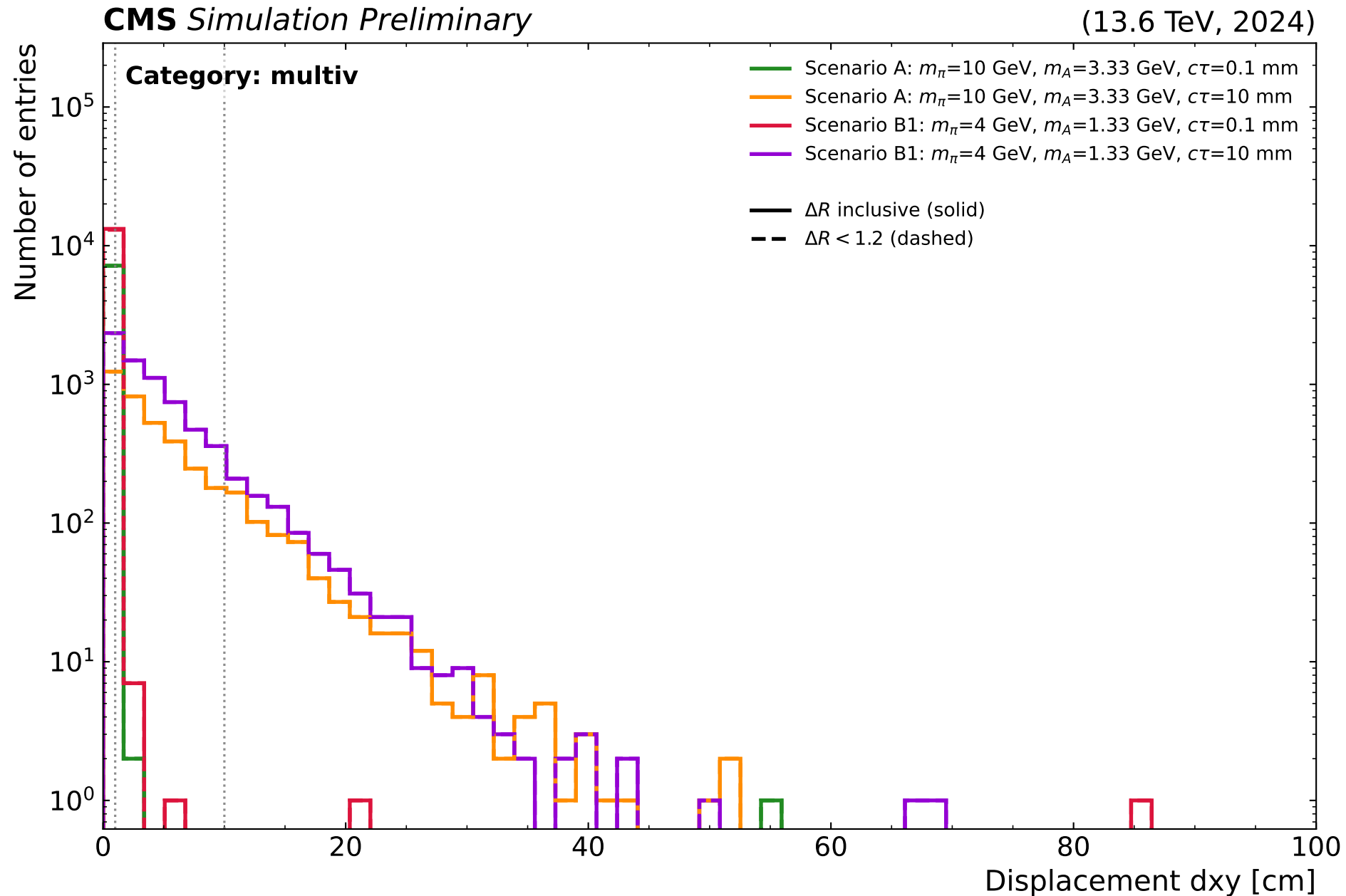
multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary

(13.6 TeV, 2024)

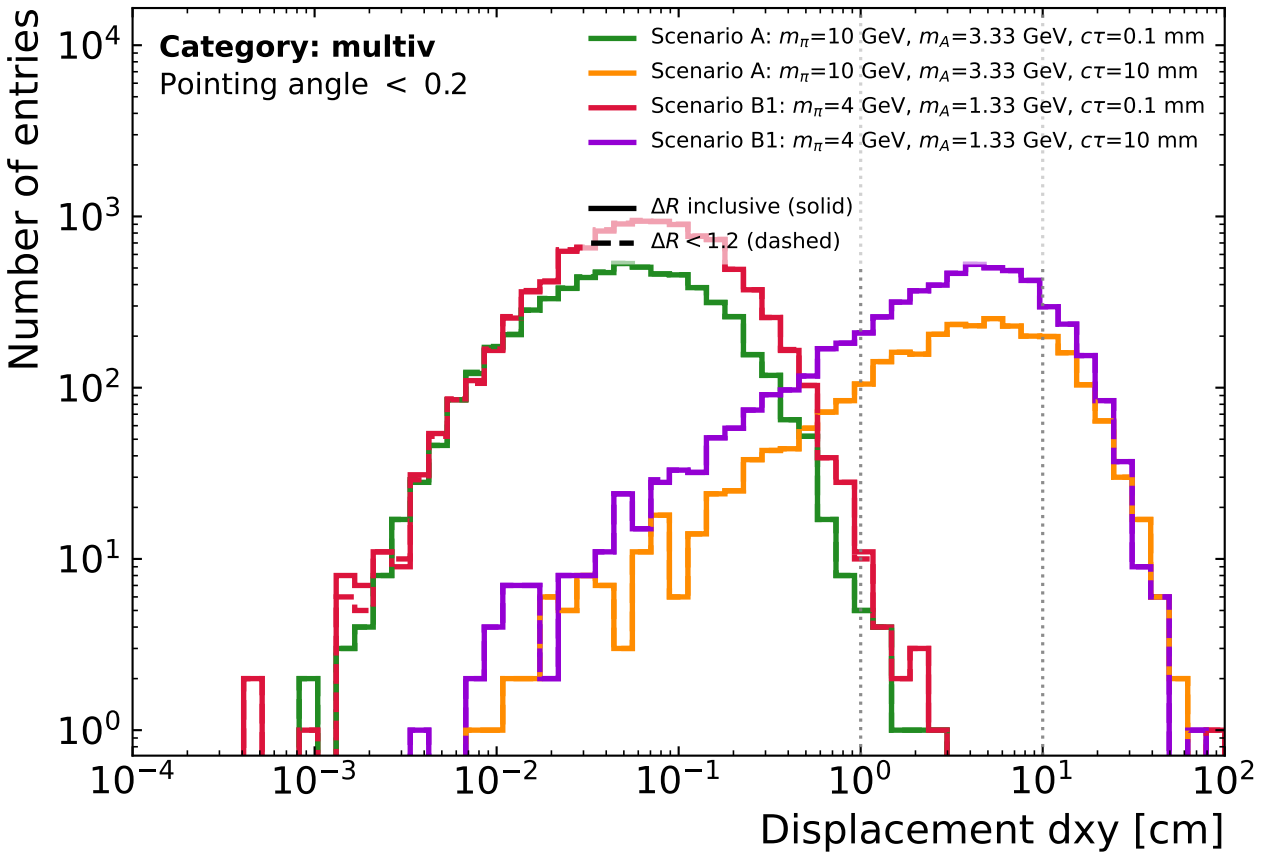


multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)



multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
 ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

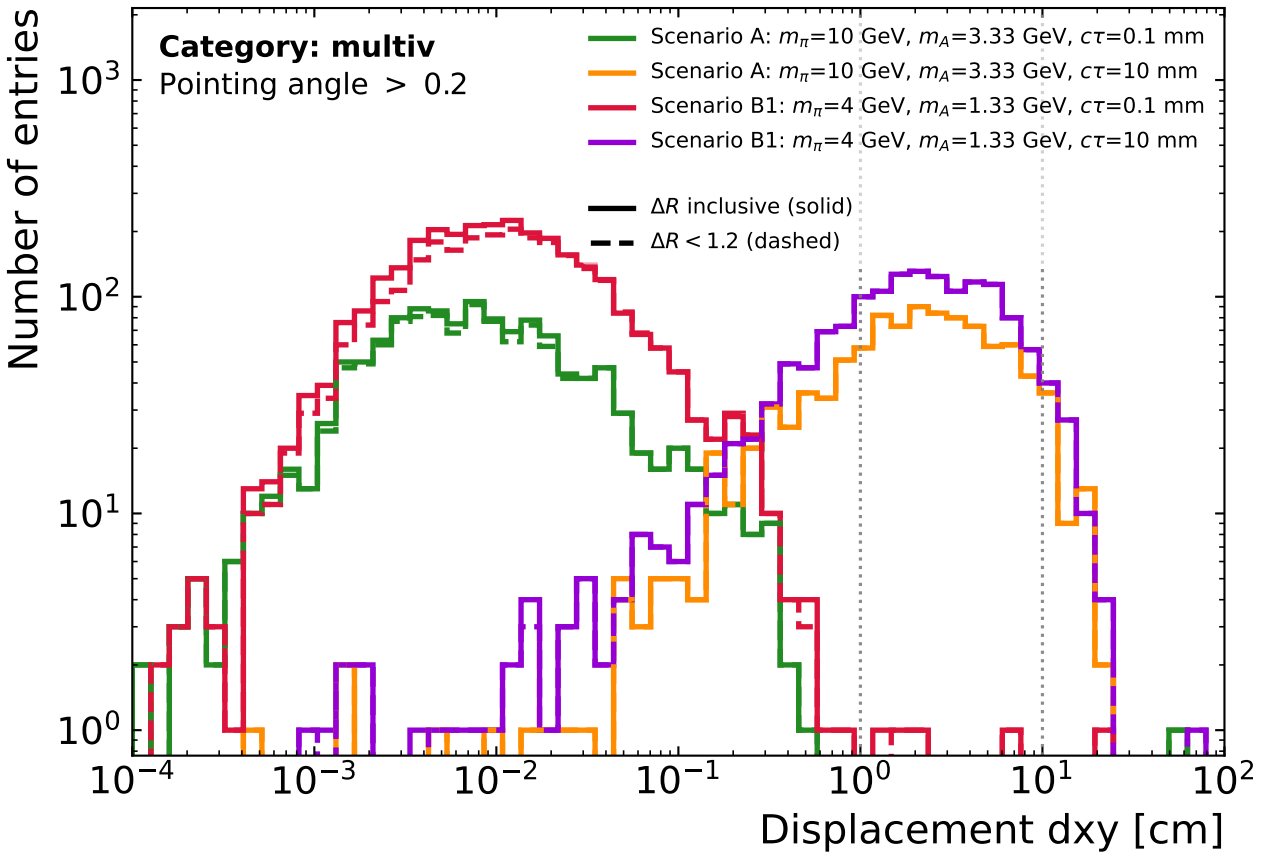
CMS Simulation Preliminary (13.6 TeV, 2024)



cell: $N(\Delta R \text{ inclusive})$ | $N(\Delta R < 1.2)$

	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	5,926 5,948	9 9	0 0	5,935 5,957
scA $c\tau=10$	501 501	1,922 1,922	547 547	2,970 2,970
scB1 $c\tau=0.1$	10,248 10,324	18 18	1 1	10,267 10,343
scB1 $c\tau=10$	1,107 1,108	3,923 3,924	761 761	5,791 5,793

CMS Simulation Preliminary (13.6 TeV, 2024)



cell: $N(\Delta R \text{ inclusive})$ | $N(\Delta R < 1.2)$

	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	1,248 1,191	0 0	1 1	1,249 1,192
scA $c\tau=10$	280 279	685 685	55 55	1,020 1,019
scB1 $c\tau=0.1$	2,962 2,680	4 3	1 1	2,967 2,684
scB1 $c\tau=10$	425 422	1,039 1,039	72 72	1,536 1,533